1. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

```
import matplotlib.pyplot as plt

data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
]

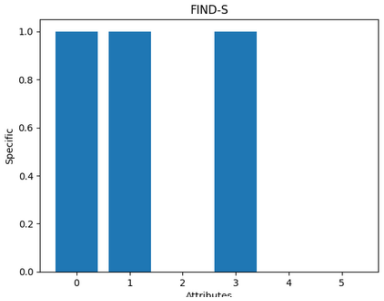
h = ['0', '0', '0', '0', '0', '0']

for row in data:
    if row[6] == 'Yes':
        for i in range(6):
            if h[i] == '0':
                h[i] = row[i]
            elif h[i] != row[i]:
                h[i] = '?'

print("Hypothesis:", h)

plt.bar(range(6), [1 if x != '?' else 0 for x in h])
plt.xlabel("Attributes")
plt.ylabel("Specific")
plt.title("FIND-S")
plt.show()
```

```
Hypothesis: ['Sunny', 'Warm', '?', 'Strong', '?', '?']
```



2. For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples

```
import matplotlib.pyplot as plt

data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
]

S = ['0', '0', '0', '0', '0', '0']

for row in data:
    if row[6] == 'Yes':
        for i in range(6):
            if S[i] == '0':
                S[i] = row[i]
            elif S[i] != row[i]:
                S[i] = '?'

print("Specific Boundary:", S)

plt.bar(range(6), [1 if x != '?' else 0 for x in S])
plt.title("Candidate Elimination")
plt.show()
```

```
Specific Boundary: ['Sunny', 'Warm', '?', 'Strong', '?', '?']
```

